CSE 270 Sports Analytics

PSS4: Networks

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```
post_soviet_countries <- c("Armenia", "Russia", "Ukraine", "Georgia", "Belarus",
                           "Azerbaijan", "Moldova", "Latvia", "Estonia")

eurovision_1 <- read.csv("eurovision.csv") %>%
  filter(Duplicate != "x")

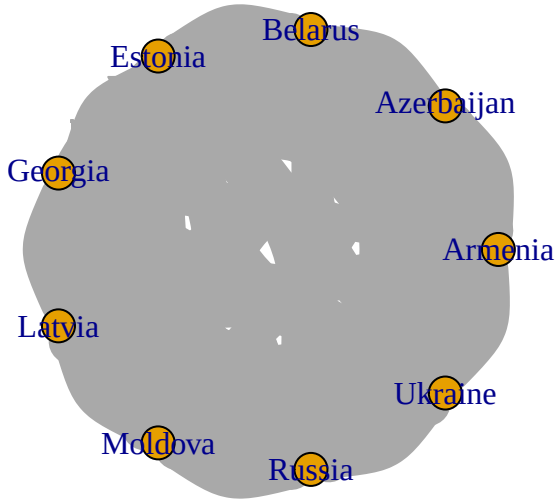
eurovision_ussr <- eurovision_1 %>%
  filter(From.country %in% post_soviet_countries,
         To.country %in% post_soviet_countries,
         From.country != To.country ) %>%
  group_by(From.country, To.country) %>%
  summarise(count = n(),
            mean_points = mean(Points),
            total_points = sum(Points)
            )


ussr_igraph <- igraph::graph_from_edgelist(as.matrix(eurovision_ussr[,1:2]), directed = T)

ussr_igraph <- set_edge_attr(ussr_igraph, "mean.points", value = eurovision_ussr$mean_points)
ussr_igraph <- set_edge_attr(ussr_igraph, "count", value = eurovision_ussr$count)
ussr_igraph <- set_edge_attr(ussr_igraph, "total.points", value = eurovision_ussr$total_points)
```

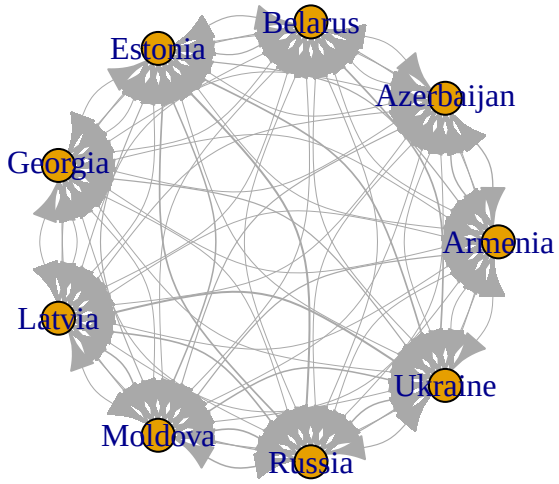
```
l <- layout_in_circle(ussr_igraph)

plot(ussr_igraph, layout = l, edge.arrow.mode = 3,
     edge.width = E(ussr_igraph)$"count",edge.curved = T,
     label = V(ussr_igraph))
```



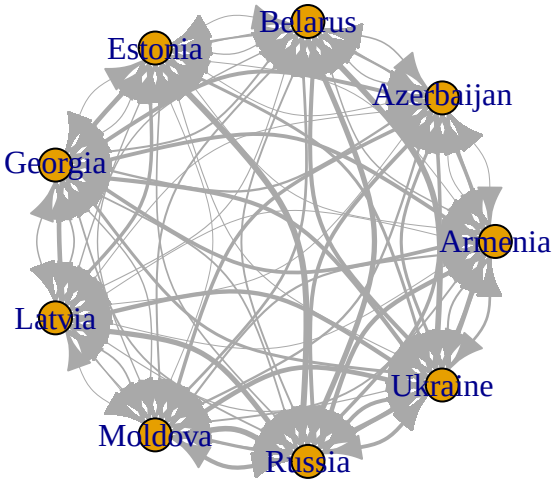

```
ussr_count <- E(ussr_igraph)$"count"
ussr_count <- (ussr_count - min(ussr_count))/
  (max(ussr_count) - min(ussr_count))
ussr_count[which(ussr_count == 0, arr.ind = T)] <- 0.0000001
ussr_igraph <- set_edge_attr(ussr_igraph, "scaled.count", value = ussr_count)


plot(ussr_igraph, layout = l, edge.arrow.mode = 3,
     edge.width = E(ussr_igraph)$"scaled.count",edge.curved = T,
     label = V(ussr_igraph))
```




```
ussr_total_points <- E(ussr_igraph)$"total.points"
ussr_total_points <- (ussr_total_points - min(ussr_total_points))/
  (max(ussr_total_points) - min(ussr_total_points))
ussr_total_points[which(ussr_total_points == 0, arr.ind = T)] <- 0.0000001
ussr_igraph <- set_edge_attr(ussr_igraph, "scaled.total.points", value = ussr_total_points)


plot(ussr_igraph, layout = l, edge.arrow.mode = 3,
     edge.width = E(ussr_igraph)$"scaled.total.points" * 3,edge.curved = T,
     label = V(ussr_igraph))
```




```
eurovision <- read.csv("eurovision.csv") %>%
  filter(Duplicate != "x", From.country != To.country ) %>%
  select(-Duplicate) %>%
  group_by(From.country, To.country)%>%
  summarise(count = n(),
          mean_points = mean(Points),
          total_points = sum(Points)
          )

evs_igraph <- igraph::graph_from_edgelist(as.matrix(eurovision[,1:2]), directed = T)
evs_igraph <- set_edge_attr(evs_igraph, "count", value = eurovision$count)
evs_igraph <- set_edge_attr(evs_igraph, "mean.points", value = eurovision$mean_points)
evs_igraph <- set_edge_attr(evs_igraph, "total_points", value = eurovision$total_points)
```

```
(density <- igraph::edge_density(evs_igraph, loops = F))
```

```
## [1] 0.8703704
```

```
(igraph::reciprocity(evs_igraph))
```

```
## [1] 0.9867523
```

```
sort(igraph::degree(evs_igraph, mode = "all", normalized = T), decreasing = T)
```

```
##              Denmark              Finland      The Netherlands
##            1.9811321            1.9811321            1.9811321
##              Austria              Belgium               France
##            1.9622642            1.9622642            1.9622642
##               Greece                Spain              Germany
##            1.9622642            1.9622642            1.9433962
##              Ireland               Norway             Portugal
##            1.9433962            1.9433962            1.9433962
##               Sweden       United Kingdom          Switzerland
##            1.9433962            1.9433962            1.9245283
##               Cyprus               Israel              Iceland
```

```
##            1.9056604            1.9056604            1.8867925
##                Malta               Turkey              Albania
##            1.8867925            1.8867925            1.8679245
##              Estonia              Hungary                Italy
##            1.8679245            1.8679245            1.8679245
##              Moldova             Slovenia              Belarus
##            1.8679245            1.8679245            1.8490566
##              Croatia              Romania               Russia
##            1.8490566            1.8490566            1.8490566
## Bosnia & Herzegovina             Bulgaria            Lithuania
##            1.8301887            1.8301887            1.8301887
##              Ukraine              Armenia     F.Y.R. Macedonia
##            1.8301887            1.8113208            1.8113208
##               Latvia               Poland               Serbia
##            1.8113208            1.8113208            1.7924528
##              Georgia           Azerbaijan       Czech Republic
##            1.7735849            1.7358491            1.7169811
##              Andorra           Montenegro           San Marino
##            1.6981132            1.6981132            1.6981132
##            Australia               Monaco             Slovakia
##            1.6415094            1.5849057            1.5283019
##  Serbia & Montenegro           Luxembourg           Yugoslavia
##            1.4528302            1.0188679            0.8679245
##              Morocco             Macedonia       The Netherands
##            0.6792453            0.6415094            0.4905660
```

```
sort(igraph::closeness(evs_igraph, mode = "all",
                       weights = E(evs_igraph)$sum.points, normalized = T), decreasing = T)
```

```
##              Austria              Denmark              Finland
##            1.0000000            1.0000000            1.0000000
##               France                Spain      The Netherlands
##            1.0000000            1.0000000            1.0000000
##              Belgium              Germany               Greece
##            0.9814815            0.9814815            0.9814815
##              Ireland               Norway             Portugal
##            0.9814815            0.9814815            0.9814815
##               Sweden       United Kingdom               Cyprus
##            0.9814815            0.9814815            0.9636364
##               Israel          Switzerland              Albania
##            0.9636364            0.9636364            0.9464286
##              Estonia              Hungary              Iceland
##            0.9464286            0.9464286            0.9464286
##                Italy                Malta              Moldova
##            0.9464286            0.9464286            0.9464286
##             Slovenia               Turkey              Armenia
##            0.9464286            0.9464286            0.9298246
##              Belarus Bosnia & Herzegovina             Bulgaria
##            0.9298246            0.9298246            0.9298246
##              Croatia            Lithuania              Romania
##            0.9298246            0.9298246            0.9298246
##               Russia              Ukraine     F.Y.R. Macedonia
##            0.9298246            0.9298246            0.9137931
##               Latvia               Poland               Serbia
```

```
##           0.9137931           0.9137931           0.9137931
##             Georgia          Azerbaijan      Czech Republic
##           0.8983051           0.8833333           0.8833333
##             Andorra           Australia          Montenegro
##           0.8688525           0.8688525           0.8688525
##          San Marino              Monaco            Slovakia
##           0.8688525           0.8281250           0.8153846
##  Serbia & Montenegro          Luxembourg      The Netherands
##           0.7910448           0.6708861           0.6625000
##          Yugoslavia           Macedonia             Morocco
##           0.6385542           0.6091954           0.6022727
```

```
sort(page_rank(evs_igraph)$vector, decreasing = T)
```

```
##             Denmark             Finland     The Netherlands
##         0.020972702         0.020972702         0.020972702
##             Belgium              Greece             Austria
##         0.020883350         0.020883350         0.020455302
##               Spain              France             Ireland
##         0.020455302         0.020455302         0.020448853
##              Norway            Portugal             Germany
##         0.020448853         0.020448853         0.020448853
##              Sweden      United Kingdom         Switzerland
##         0.020448853         0.020448853         0.020359530
##              Cyprus              Israel              Turkey
##         0.020027039         0.020027039         0.020012053
##             Iceland               Malta             Albania
##         0.019937745         0.019937745         0.019776096
##             Moldova             Estonia             Hungary
##         0.019776096         0.019776096         0.019776096
##               Italy             Belarus              Russia
##         0.019761868         0.019686833         0.019686833
##             Romania             Armenia            Slovenia
##         0.019686833         0.019672559         0.019633457
##             Croatia           Lithuania             Ukraine
##         0.019544194         0.019252367         0.019252367
##            Bulgaria Bosnia & Herzegovina              Latvia
##         0.019252367         0.019203506         0.019163135
##    F.Y.R. Macedonia              Poland              Serbia
##         0.019163135         0.019163135         0.019083737
##             Georgia          Azerbaijan      Czech Republic
##         0.018994538         0.018470889         0.018218911
##             Andorra           Montenegro          San Marino
##         0.018130643         0.018129782         0.018129782
##           Australia              Monaco            Slovakia
##         0.017533775         0.017183367         0.016398350
##  Serbia & Montenegro          Luxembourg          Yugoslavia
##         0.015715809         0.012094217         0.010646597
##           Macedonia             Morocco      The Netherands
##         0.009279834         0.008940036         0.002777778
```

```
undirected <- as.undirected(evs_igraph, mode = c("collapse"))
com <- fastgreedy.community(undirected)
sizes(com)
```

```
## Community sizes
##  1  2
## 29 25
```

```
plot(com, evs_igraph)
```

